

Evidence Trails for Agentic Research Workflows

Abstract

Automated research workflows need reviewable evidence: scenario inputs, tool calls, intermediate decisions, touched files, failure labels, recovery behavior, and final outputs. This manuscript outline treats agent runs as experimental records that support comparison across systems and prompts.

Methodology

The evaluation record stores the initial task, tool calls, touched files, intermediate outputs, failure labels, recovery actions, and final diff. Runs are compared at the scenario level, with enough structure to inspect both successful and failed attempts.

Results

The manuscript defines an evidence trail for agentic research workflows: a run can be audited, replayed at the level of decisions, and compared across prompts, models, and tool environments.